MIC regression

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Preamble

Fit regression models

Complete MIC data (NOS)

```
# apply log to the mics (mix of NOS and 90):
dat_nos_only <- dat_nos_only %>%
  mutate(mic_logged = log2(micnos_1))

# fit a model weighted by inverse sqrt sample sizes to determine residual errors:
# weights in full model
mod_init <- lm(mic_logged ~ isolate_yr, dat_nos_only, weights = 1/sqrt(count))
summary(mod_init)
```

```
##
## Call:
## lm(formula = mic_logged ~ isolate_yr, data = dat_nos_only, weights = 1/sqrt(count))
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -2.98965 -0.39038 -0.00134  0.01035  2.45339
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -5.2795876 13.0483883  -0.405    0.686
## isolate_yr  -0.0003462  0.0065127  -0.053    0.958
##
## Residual standard error: 0.6195 on 170 degrees of freedom
## Multiple R-squared:  1.662e-05,  Adjusted R-squared:  -0.005866
## F-statistic: 0.002826 on 1 and 170 DF,  p-value: 0.9577
```

```
# setting weight to median absolute residuals
wts <- data.frame(isolate_yr = dat_nos_only$isolate_yr,
                  resid = mod_init$residuals) %>%
  group_by(isolate_yr) %>%
  summarise(wt = median(abs(resid)))

dat_nos_only <- dat_nos_only %>%
  left_join(wts)
```

```
## Joining with `by = join_by(isolate_yr)`
```

```r
mod_weighted_nos_only <- lm(mic_logged ~ isolate_yr, dat_nos_only, weights = wt)
confint(mod_weighted_nos_only)
```

```
##                    2.5 %     97.5 %
## (Intercept) -53.81826266 0.17663911
## isolate_yr   -0.00315681 0.02384291
```

```r
summary(mod_weighted_nos_only)
```

```
##
## Call:
## lm(formula = mic_logged ~ isolate_yr, data = dat_nos_only, weights = wt)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -2.87548 -0.58970  0.01371  0.12923  2.96707
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -26.820812  13.676404  -1.961   0.0515 .
## isolate_yr    0.010343   0.006839   1.512   0.1323
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8733 on 170 degrees of freedom
## Multiple R-squared:  0.01328,    Adjusted R-squared:  0.007472
## F-statistic: 2.287 on 1 and 170 DF,  p-value: 0.1323
```

```r
# this is slope on log-scale:
coefficients(mod_weighted_nos_only)[2]
```

```
## isolate_yr
## 0.01034305
```

```r
2**(coefficients(mod_weighted_nos_only)[2])
```

```
## isolate_yr
##   1.007195
```

```r
2**confint(mod_weighted_nos_only)[2,]
```

```
##     2.5 %    97.5 %
## 0.9978143 1.0166640
```

```r
dat_nos_only %>%
  # clarify this is right:
  group_by(author, pub_yr) %>%
  summarise(n = n()) %>%
  as.data.frame()
```

Adding random effect on study

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by author and pub_yr.
## i Output is grouped by author.
## i Use `summarise(.groups = "drop_last")` to silence this message.
```

```
## i Use `summarise(.by = c(author, pub_yr))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```
##           author pub_yr  n
## 1   Abd El Ghany   2015  3
## 2   Abd El-Ghany   2015  1
## 3         Amusin   2026  4
## 4          Arhin   2009  3
## 5        Benouda   2009  3
## 6         Canton   2002 33
## 7        Cellesi   1996  3
## 8          Cohen   1996  4
## 9        Feldman   1987  1
## 10        Friaes   2025  1
## 11        Fujita   1994  1
## 12         Haddy   1982  3
## 13         Ikeda   2021  3
## 14 Kanellopoulou   1995  1
## 15           Kim   1985  2
## 16   Kriebernegg   1998  1
## 17        Musser   2020 54
## 18      Nakajima   1979  1
## 19      Nakajima   1982  3
## 20      Nakajima   1983  4
## 21      Reynolds   2025  6
## 22        Sakata   2013  2
## 23     Smeesters   2010  5
## 24       Southon   2020  7
## 25      Soyletir   2016  7
## 26    Stillerman   1970  3
## 27    Stillerman   1973  4
## 28         Tappe   2011  1
## 29    Torumkuney   2016  3
## 30       Vannice   2020  2
## 31       Varaldo   1999  3
```

```
X_LOWER <- 1965

# note misspelling of Abd El[ |-]Ghany

library(lme4)
```

```
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
```

```
dat_nos_only_uncounted <- dat_nos_only %>%
  mutate(study = factor(paste(author, pub_yr, sep = ","))) %>%
  uncount(count)

mod_fixed <- lm(mic_logged ~ isolate_yr, dat_nos_only_uncounted)
tmp <- 2**confint(mod_fixed)
```

```r
tmp[2, 1:2]
```

```
##     2.5 %    97.5 %
## 0.9966422 0.9981378
```

```r
# we don't have multiple observations in the same study across multiple years - do we gain
# much structure from random effect on study?
mod_mixed <- lmer(mic_logged ~ isolate_yr + (1|study), dat_nos_only_uncounted)
mod_mixed <- lmer(mic_logged ~ isolate_yr + (1|country), dat_nos_only_uncounted)

summary(mod_mixed)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: mic_logged ~ isolate_yr + (1 | country)
##    Data: dat_nos_only_uncounted
##
## REML criterion at convergence: 13895.2
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -6.2612 -0.0213  0.1076  0.3098  6.1028
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  country  (Intercept) 0.6583   0.8113
##  Residual             0.2518   0.5018
## Number of obs: 9401, groups:  country, 33
##
## Fixed effects:
##               Estimate Std. Error t value
## (Intercept) -1.483e+01  1.114e+00 -13.314
## isolate_yr   4.135e-03  5.523e-04   7.488
##
## Correlation of Fixed Effects:
##            (Intr)
## isolate_yr -0.992
```

```r
# this is what we want in the text:
tmp <- 2**confint(mod_mixed)
```

```
## Computing profile confidence intervals ...
```

```r
# ... and now we're slightly positive
tmp[4, 1:2]
```

```
##    2.5 %   97.5 %
## 1.002119 1.003624
```

```r
# dropping code for plot up here:
newdata <- data.frame(isolate_yr = seq(X_LOWER, max(dat_nos_only$isolate_yr) + 2),
                      study = "NA,NA", country = "NA")

fixed_eff_pred <- data.frame(y = 2**predict(mod_fixed, newdata),
                             x = newdata$isolate_yr)

# bootstrap to get approximate CI
boot_ci <- bootMer(mod_mixed,
```

```
                  nsim = 100,
                  FUN = function(x){
                    predict(x, newdata = newdata, re.form = NA)
                    # ignoring random effect in prediction
                  })

mixed_eff_pred <- apply(boot_ci$t, 2, function(x){quantile(x, probs = c(0.025, 0.5, 0.975))}) %>%
  t() %>%
  as.data.frame() %>%
  setNames(nm = c("lwr", "fit", "upr")) %>%
  bind_cols(newdata %>% dplyr::select(isolate_yr))

YLIM <- range(c(dat_agg$mic_90_agg,
                dat_nos_only$micnos_1))
Y_LOWER <- log2(0.0018)
SIZE_LIMS <- range(c(dat_agg$count,
                     dat_nos_only$count))
yticks <- c(0, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.12)

p1 <- ggplot(dat_nos_only %>%
             filter(!is.na(micnos_1)) %>%
             group_by(isolate_yr, micnos_1, who_name) %>%
             summarise(count = sum(count)) %>%
             mutate(who_name = factor(who_name, levels = c("Africa",
                                                           "Americas",
                                                           "Eastern Mediterranean",
                                                           "Europe",
                                                           "Southeast Asia",
                                                           "Western Pacific",
                                                           "Not assigned"))) %>%
         arrange(desc(count))) +
  geom_line(data = fixed_eff_pred, aes(x = x, y = y)) +
  geom_ribbon(data = mixed_eff_pred, aes(ymin = 2**lwr, ymax = 2**upr, x = isolate_yr),
              alpha = 0.5, fill = "grey", colour = NA) +
  geom_line(data = mixed_eff_pred, aes(x = isolate_yr, y = 2**fit),
            colour = "darkgrey", linewidth = 0.8) +
  geom_point(aes(x = isolate_yr, y = micnos_1,
                 col = who_name,
                 size = count), alpha = 0.5) +
  scale_color_discrete(name = "WHO Region", type = pal, drop = FALSE) +
  scale_size_continuous(range = c(1,10), name = "Number of isolates",
                        breaks = c(1, 10, 100, 1000), limits = SIZE_LIMS) +
  scale_y_continuous(breaks = yticks, minor_breaks = NULL, name = "MIC") +
  scale_x_continuous(limits = c(X_LOWER, 2022), expand = c(0,0),
                     breaks = seq(1970, 2010, 10), minor_breaks = NULL) +
  labs(x = "Year", ylab = "MIC") +
  coord_trans(y = "log2", ylim = YLIM) +
  theme_bw()
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by isolate_yr, micnos_1, and who_name.
## i Output is grouped by isolate_yr and micnos_1.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(isolate_yr, micnos_1, who_name))` for per-operation
```

```
##   grouping (`?dplyr::dplyr_by`) instead.
```

```
## Warning: `coord_trans()` was deprecated in ggplot2 4.0.0.
## i Please use `coord_transform()` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

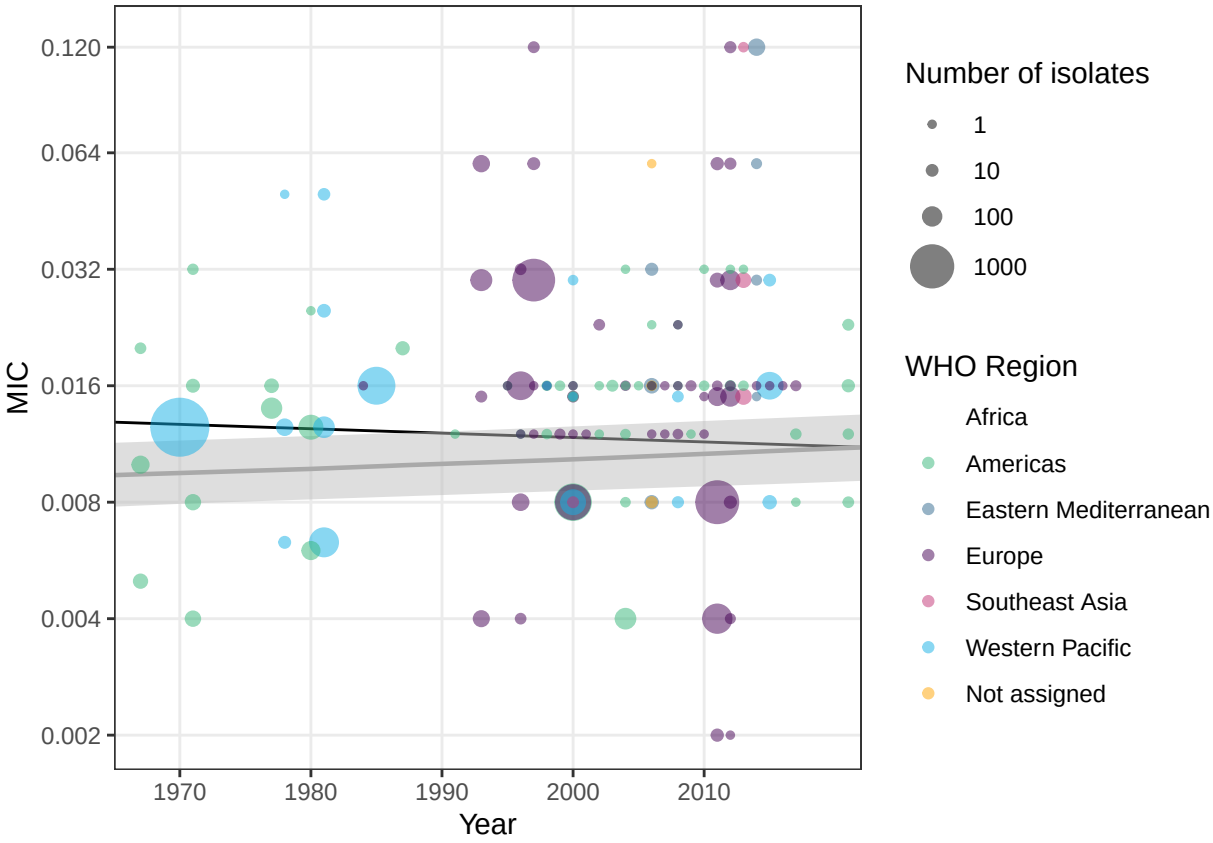
```
# okey well there's something weird happening here
p1
```

```
## Ignoring unknown labels:
## * ylab : "MIC"
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_line()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_ribbon()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_line()`).
```



All studies: MIC90

```
# apply log to the mics (mix of NOS and 90):
dat_agg <- dat_agg %>%
  mutate(mic_logged = log2(mic_90_agg))

# shouldn't these be one per study?
```

```r
dat_agg %>%
  filter(!is.na(mic_logged)) %>%
  group_by(author, pub_yr, country) %>%
  summarise(n = n()) %>%
  filter(n > 1)
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by author, pub_yr, and country.
## i Output is grouped by author and pub_yr.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(author, pub_yr, country))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```
## # A tibble: 23 x 4
## # Groups:   author, pub_yr [9]
##    author     pub_yr country                          n
##    <chr>       <int> <chr>                        <int>
##  1 Bass         1994 Japan                            2
##  2 Buxbaum      2004 Austria                          4
##  3 Chazan       2015 Israel                           2
##  4 Coonan       1994 United States Of America (The)   2
##  5 Inoue        2006 Japan                            2
##  6 Macris       1998 United States Of America (The)   8
##  7 Marchese     2004 Italy                            2
##  8 d'Humières   2012 France                           2
##  9 <NA>           NA Australia                        2
## 10 <NA>           NA Austria                          2
## # i 13 more rows
```

```r
# fit a model weighted by inverse sqrt sample sizes to determine residual errors:
# weights in full model
mod_init = lm(mic_logged ~ isolate_yr, dat_agg, weights = 1/sqrt(count))
summary(mod_init)
```

```
##
## Call:
## lm(formula = mic_logged ~ isolate_yr, data = dat_agg, weights = 1/sqrt(count))
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -0.72980 -0.20952 -0.00096  0.11378  1.40323
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.943212   6.839889  -0.284    0.777
## isolate_yr  -0.002053   0.003418  -0.601    0.549
##
## Residual standard error: 0.3147 on 216 degrees of freedom
## Multiple R-squared:  0.001668,   Adjusted R-squared:  -0.002954
## F-statistic: 0.3608 on 1 and 216 DF,  p-value: 0.5487
```

```r
# setting weight to median absolute residuals
wts <- data.frame(isolate_yr = dat_agg$isolate_yr,
                  resid = mod_init$residuals) %>%
  group_by(isolate_yr) %>%
  summarise(wt = median(abs(resid)))
```

```r
dat_agg <- dat_agg %>%
  left_join(wts)
```

```
## Joining with `by = join_by(isolate_yr)`
```

```r
mod_weighted_agg <- lm(mic_logged ~ isolate_yr, dat_agg, weights = wt)
confint(mod_weighted_agg)
```

```
##                    2.5 %      97.5 %
## (Intercept) -28.82470320 19.03996449
## isolate_yr   -0.01250746  0.01143648
```

```r
summary(mod_weighted_agg)
```

```
##
## Call:
## lm(formula = mic_logged ~ isolate_yr, data = dat_agg, weights = wt)
##
## Weighted Residuals:
##     Min      1Q  Median      3Q     Max
## -2.7877 -0.3851 -0.0055  0.1692  3.4000
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -4.8923694 12.1421832  -0.403    0.687
## isolate_yr  -0.0005355  0.0060740  -0.088    0.930
##
## Residual standard error: 0.776 on 216 degrees of freedom
## Multiple R-squared:  3.598e-05,  Adjusted R-squared:  -0.004593
## F-statistic: 0.007772 on 1 and 216 DF,  p-value: 0.9298
```

```r
# this is slope on log-scale:
coefficients(mod_weighted_agg)[2]
```

```
##    isolate_yr
## -0.0005354911
```

```r
2**(coefficients(mod_weighted_agg)[2])
```

```
## isolate_yr
##  0.9996289
```

```r
2**confint(mod_weighted_agg)[2,]
```

```
##    2.5 %   97.5 %
## 0.991368 1.007959
```

Now for some plots

Common attributes:

```r
YLIM <- range(c(dat_agg$mic_90_agg, dat_nos_only$micnos_1))
YLIM <- c(0.0018, 0.128)
Y_LOWER <- log2(0.0018)
SIZE_LIMS <- range(c(dat_agg$count, dat_nos_only$count))
yticks <- c(0, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.12)
```

Complete MIC data (NOS only)

```
X_LOWER <- 1965

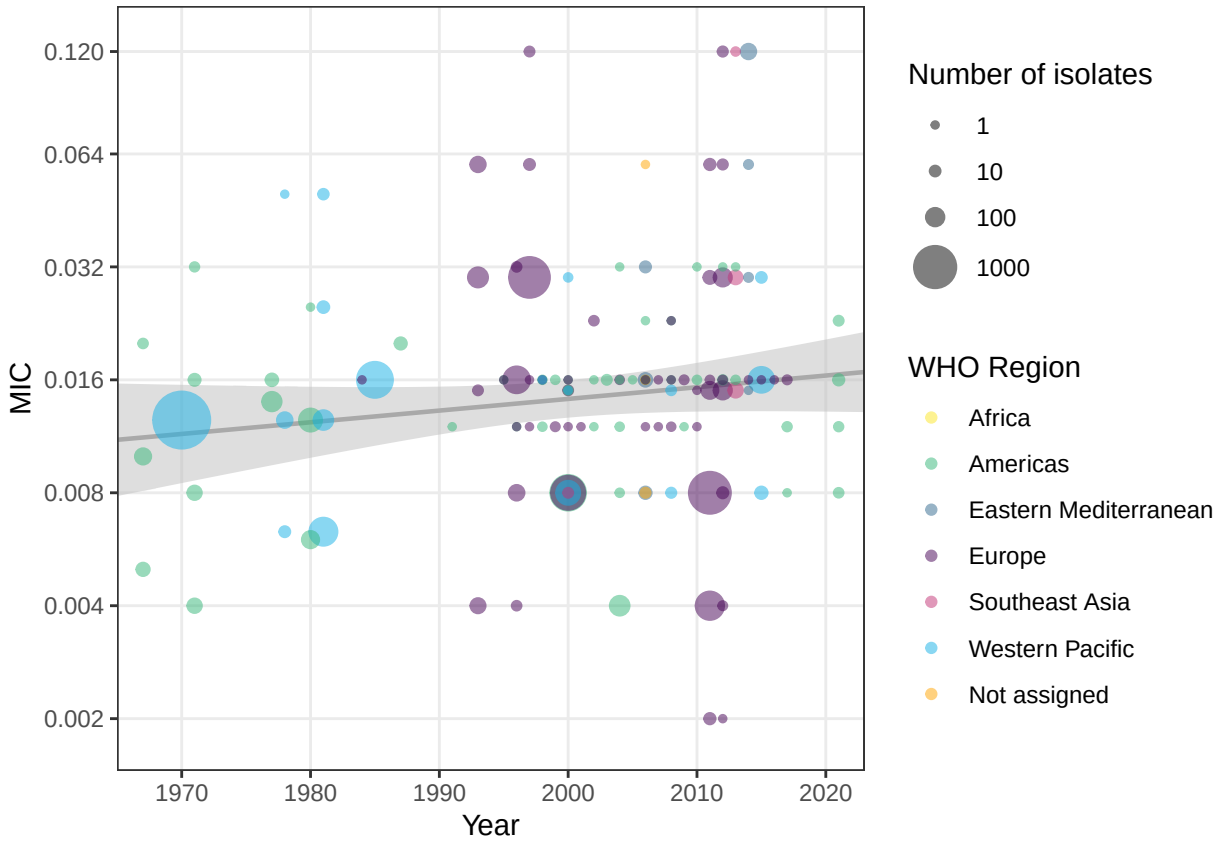
# here comes the model again:
newdata <- data.frame(isolate_yr = seq(X_LOWER, max(dat_nos_only$isolate_yr) + 2))
ci <- predict(mod_weighted_nos_only,
                newdata = newdata,
                interval = "confidence") %>%
  as.data.frame() %>%
  mutate(lwr = ifelse(lwr < Y_LOWER, Y_LOWER, lwr),
         isolate_yr = seq(X_LOWER, max(dat_nos_only$isolate_yr) + 2))

p1 <- ggplot(dat_nos_only %>%
             filter(!is.na(micnos_1)) %>%
             group_by(isolate_yr, micnos_1, who_name) %>%
             summarise(count = sum(count)) %>%
             mutate(who_name = factor(who_name, levels = c("Africa",
                                                           "Americas",
                                                           "Eastern Mediterranean",
                                                           "Europe",
                                                           "Southeast Asia",
                                                           "Western Pacific",
                                                           "Not assigned"))) %>%
         arrange(desc(count))) +
  geom_ribbon(data = ci, aes(ymin = 2**lwr, ymax = 2**upr, x = isolate_yr),
              alpha = 0.5, fill = "grey", colour = NA) +
  geom_line(data = ci, aes(x = isolate_yr, y = 2**fit),
            colour = "darkgrey", linewidth = 0.8) +
  geom_point(aes(x = isolate_yr, y = micnos_1,
                 col = who_name,
                 size = count), alpha = 0.5,
             show.legend = TRUE) +
  scale_color_manual(name = "WHO Region", values = pal, drop = FALSE) +
  scale_size_continuous(range = c(1,10), name = "Number of isolates",
                        breaks = c(1, 10, 100, 1000), limits = SIZE_LIMS) +
  scale_y_continuous(breaks = yticks, minor_breaks = NULL, name = "MIC") +
  scale_x_continuous(limits = c(X_LOWER, 2023), expand = c(0,0),
                     breaks = seq(1970, 2020, 10), minor_breaks = NULL) +
  labs(x = "Year", ylab = "MIC") +
  coord_trans(y = "log2", ylim = YLIM) +
  theme_bw()
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by isolate_yr, micnos_1, and who_name.
## i Output is grouped by isolate_yr and micnos_1.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(isolate_yr, micnos_1, who_name))` for per-operation
##   grouping (`?dplyr::dplyr_by`) instead.
```

```
p1
```

```
## Ignoring unknown labels:
## * ylab : "MIC"
```



All studies: MIC90

```
X_LOWER <- 1927

# here comes the model again:
newdata <- data.frame(isolate_yr = seq(X_LOWER, max(dat_agg$isolate_yr) + 2))
ci <- predict(mod_weighted_agg,
                newdata = newdata,
                interval = "confidence") %>%
  as.data.frame() %>%
  mutate(lwr = ifelse(lwr < Y_LOWER, Y_LOWER, lwr),
         isolate_yr = seq(X_LOWER, max(dat_agg$isolate_yr) + 2))

p2 <- ggplot(dat_agg %>%
             filter(!is.na(mic_90_agg)) %>%
             group_by(isolate_yr, mic_90_agg, who_name) %>%
             summarise(count = sum(count)) %>%
             mutate(who_name = factor(who_name, levels = c("Africa",
                                                           "Americas",
                                                           "Eastern Mediterranean",
                                                           "Europe",
                                                           "Southeast Asia",
                                                           "Western Pacific",
                                                           "Not assigned"))) %>%
         arrange(desc(count))) +
  geom_ribbon(data = ci, aes(ymin = 2**lwr, ymax = 2**upr, x = isolate_yr),
```

```
              alpha = 0.5, fill = "grey", colour = NA) +
  geom_line(data = ci, aes(x = isolate_yr, y = 2**fit),
            colour = "darkgrey", linewidth = 0.8) +
  geom_point(aes(x = isolate_yr, y = mic_90_agg,
                 col = who_name,
                 size = count), alpha = 0.5) +
  scale_color_discrete(name = "WHO Region",
                       type = pal,
                       guide = guide_legend(override.aes = list(size = 5),
                                            ncol = 2),
                       drop = FALSE) +
  scale_size_continuous(range = c(1,10), name = "Number of isolates",
                        breaks = c(1, 10, 100, 1000), limits = SIZE_LIMS) +
  scale_y_continuous(breaks = yticks, minor_breaks = NULL) +
  scale_x_continuous(limits = c(X_LOWER, 2023), expand = c(0,0),
                     breaks = seq(1930, 2020, 10), minor_breaks = NULL) +
  xlab("Year") +
  ylab("MIC90") +
  coord_trans(y = "log2", ylim = YLIM) +
  theme_bw() +
  theme(
    legend.box = "horizontal",
    legend.box.just = "top",
    plot.background = element_rect(fill = "transparent", colour = "transparent")
  )
```

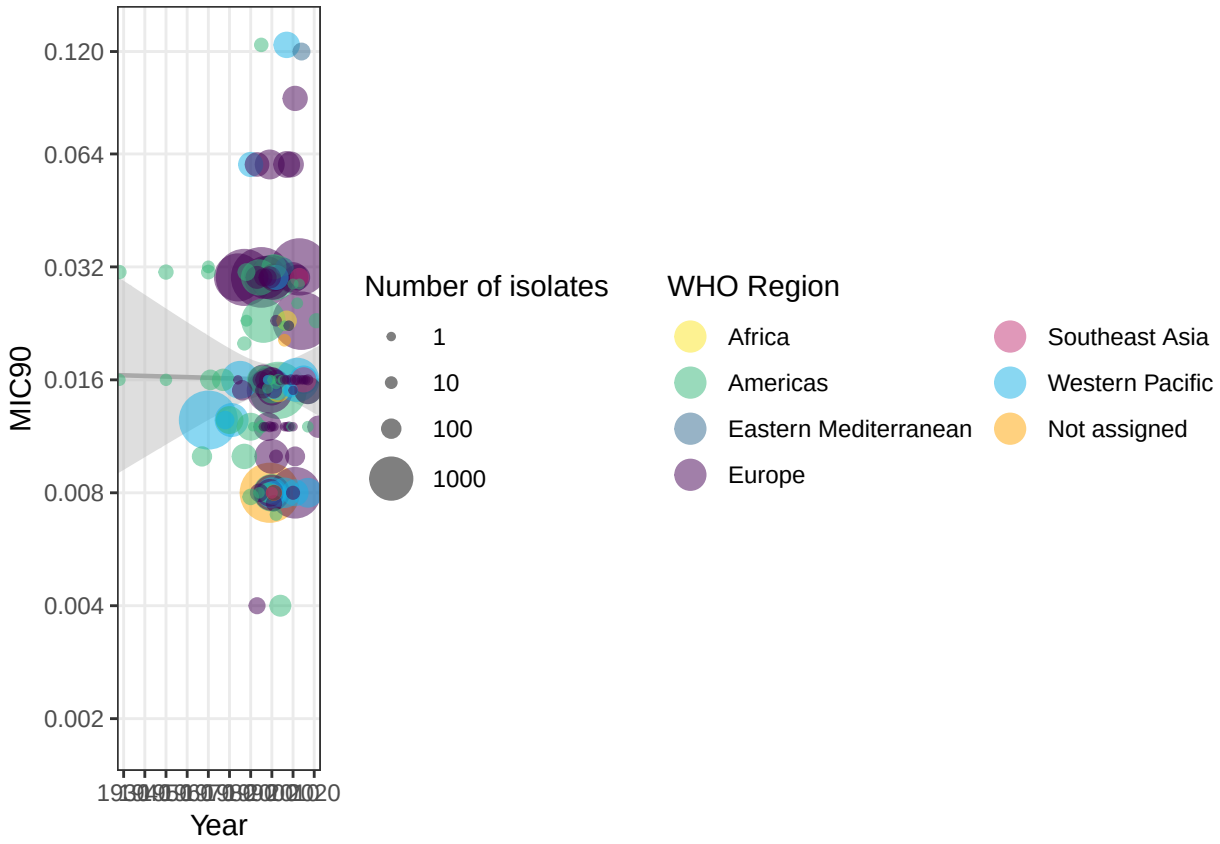
```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by isolate_yr, mic_90_agg, and who_name.
## i Output is grouped by isolate_yr and mic_90_agg.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(isolate_yr, mic_90_agg, who_name))` for per-operation
##   grouping (`?dplyr::dplyr_by`) instead.
```

```
p2
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_ribbon()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_line()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```



Putting it all together:

```
p <- plot_grid(p2 + theme(legend.position = "none",
                          plot.margin = margin(t = 20, r = 0, b = 0, l = 5, unit = "pt")),
               p1 + theme(legend.position = "none",
                          plot.margin = margin(t = 20, r = 0, b = 0, l = 5, unit = "pt")),
               labels = c("(a) MIC90: all studies",
                          "(b) Studies documenting MIC for all isolates"),
               label_fontface = "plain",
               rel_widths = c(0.6, 0.4),
               label_x = c(0.15, 0.1), hjust = 0, label_size = 12)
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_ribbon()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_line()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```

```
## Ignoring unknown labels:
## * ylab : "MIC"
```

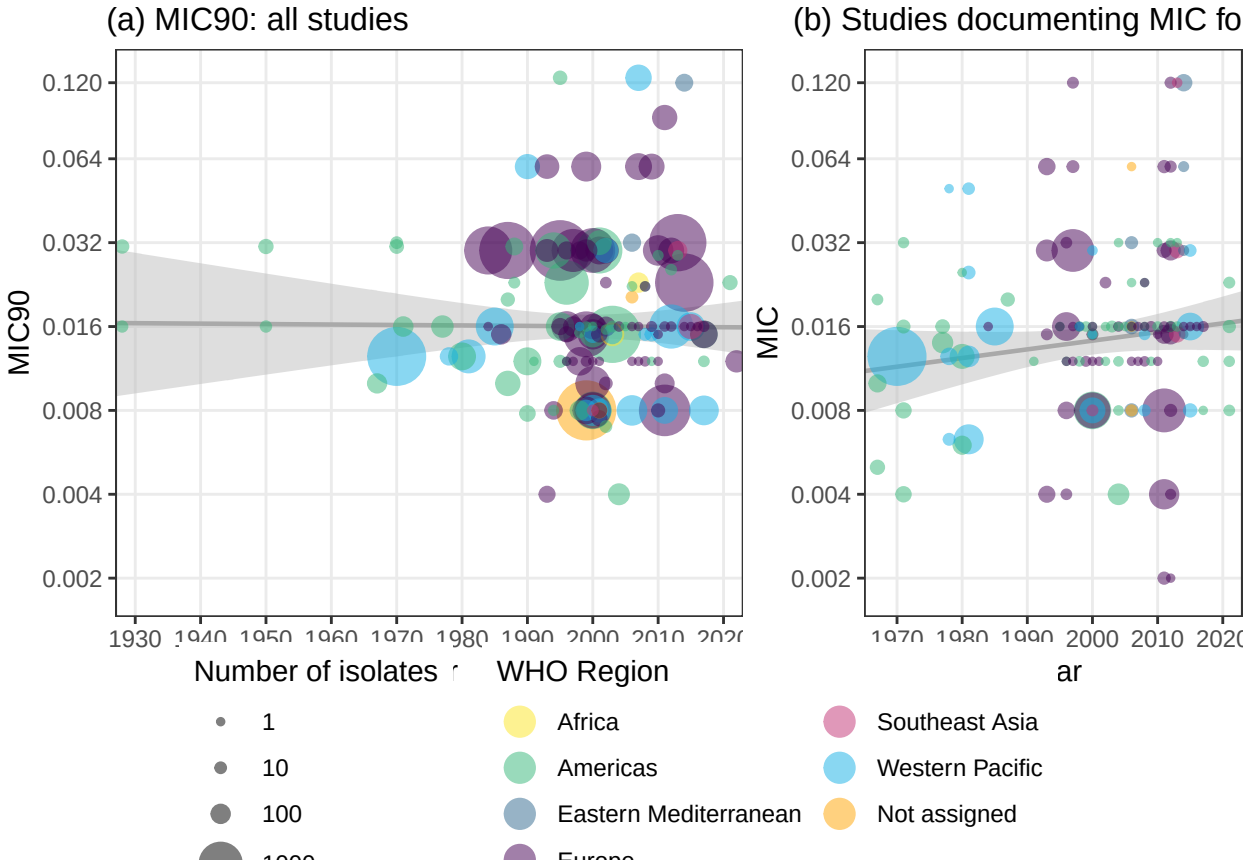
```
plot_grid(p, get_legend(p2), ncol = 1, rel_heights = c(0.8, 0.2))
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_ribbon()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_line()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```

```
## Warning in get_plot_component(plot, "guide-box"): Multiple components found;
## returning the first one. To return all, use `return_all = TRUE`.
```




```
ggsave("figs/MIC_regression_twice.png", height = 7, width = 9, bg = "transparent")
ggsave("figs/MIC_regression_twice.svg", height = 7, width = 9, bg = "transparent")
```

LH coming back, making times match vertically.

```
p <- plot_grid(
          p1 + theme(legend.position = c(-0.5, 0.5),
                     plot.margin = margin(t = 20, r = 0, b = 0, l = 215,
                                          unit = "pt")),
          p2 + theme(legend.position = "none",
                     plot.margin = margin(t = 20, r = 0, b = 0, l = 5,
                                          unit = "pt")),
          labels = c("(a) Trend based on individual MICs",
                     "(b) Trend based on summarised MICs"),
          label_fontface = "plain",
          nrow = 2,
          #label_x = c(0.07, 0.45),
          label_x = c(0.45, 0.07),
          hjust = 0, label_size = 12)
```

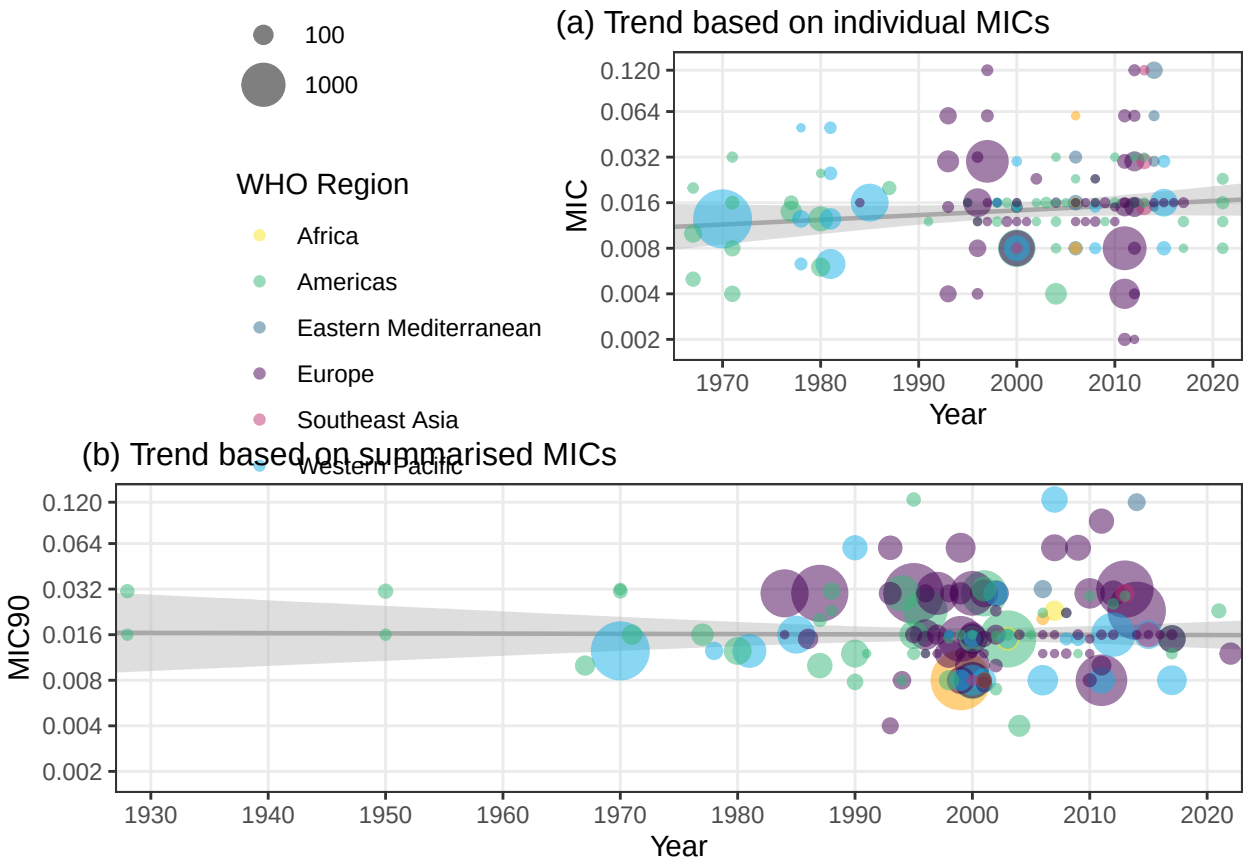
```
## Ignoring unknown labels:
## * ylab : "MIC"
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_ribbon()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_line()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```

```
p
```




```
ggsave("figs/MIC_regression_twice_vertical.png", height = 9, width = 8, bg = "white")
ggsave("figs/MIC_regression_twice_vertical.svg", height = 9, width = 8, bg = "white")
```

TODO - report Rsq

Total isolates by MIC90/raw MIC

Figure 3: what are the spreads of the two subsets? NOTE: this calculation of counts now superseded: was previously not including summarised indiv MICs in histogram of MIC90s, which doesn't make too much and makes less sense when showing these histograms next to the regression scatter plots.

```
sum(dat_raw$count)

hist_dat <- dat_raw %>% # from Rhys' aggregate file
  filter(!is.na(mic90_1)) %>%
  bind_rows(dat_nos_only) %>% # NOS only
  pivot_longer(cols = c(micnos_1, mic90_1), names_to = "type") %>%
  filter(!is.na(value)) %>%
```

```
  mutate(value = round(value, digits = 4)) %>%
  summarise(count = sum(count), .by = c(value, type)) %>%
  mutate(logmic = log2(value)) %>%
  mutate(type = ifelse(type == "mic90_1",
                       "Only summarised MIC reported (MIC90)",
                       "Individual MIC reported"))

hist_dat %>%
  summarise(.by = type, count = sum(count))

x_breaks = c(0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.12)

xlim = c(0.0019, 0.15)

# for labels on bars:
to_lab <- hist_dat %>%
  # only highest ones:
  filter((type == "Only summarised MIC reported (MIC90)" & count > 2000) |
           (type == "Individual MIC reported" & count > 1000)) %>%
  mutate(nudge_yy = ifelse(type == "Individual MIC reported",
                           1200, 300),
         countstr = format(count, big.mark = ","),
         mic = ifelse(type == "Individual MIC reported", "MIC", "MIC90"),
         hjustt = ifelse(count %in% c(2727, 2100, 3097), 0, 0.5),
         vjustt = ifelse(count %in% c(2100), 0, 0.5),
         lab = ifelse(count %in% c(2727, 3097, 2030),
                      paste0(mic, ":\n", value, "\n(n = ", countstr, ")"),
                      paste0(mic, ": ", value, "\n(n = ", countstr, ")")))



p1 <- ggplot(data = hist_dat %>%
               filter(type == "Only summarised MIC reported (MIC90)")) +
  geom_bar(aes(x = logmic, y = count,
           fill = type),
           alpha = 0.6,
           stat = "identity",
           width = 0.05) +
  # here are labels on top of higher bars:
  geom_text(aes(x = logmic, y = count, label = lab,
                hjust = hjustt),
            data = to_lab %>%
              filter(type == "Only summarised MIC reported (MIC90)"),
             nudge_y = 1500,
            vjust = 1,
            size = 2.5) +
  facet_wrap(~ type, ncol = 1, scales = "free_y") +
  scale_x_continuous(breaks = log2(x_breaks),
                     labels = x_breaks,
                     limits = log2(xlim)) +
  scale_fill_manual(values = pal[2]) +
  ylab("") +
  xlab("MIC90") +
```

```
  theme_bw() +
  theme(legend.position = "none")

p2 <- ggplot(data = hist_dat %>%
               filter(type == "Individual MIC reported")) +
  geom_bar(aes(x = logmic, y = count,
           fill = type),
           alpha = 0.6,
           stat = "identity",
           width = 0.05) +
  # here are labels on top of higher bars:
  geom_text(aes(x = logmic, y = count, label = lab),
            data = to_lab %>%
               filter(type == "Individual MIC reported"),
             nudge_y = 400,
            vjust = 1,
            size = 2.5) +
  facet_wrap(~ type, ncol = 1, scales = "free_y") +
  scale_x_continuous(breaks = log2(x_breaks),
                     labels = x_breaks,
                     limits = log2(xlim)) +
  scale_fill_manual(values = pal[3]) +
  ylab("") +
  xlab("MIC") +
  theme_bw() +
  theme(legend.position = "none")

p <- plot_grid(p2, p1, ncol = 1)

ggdraw(p) +
  draw_label("Number of isolates", x = 0, y = 0.5, angle = 90,
             vjust = 1.5, size = 12)

ggsave("figs/all_mics.png", height = 6, width = 6)
ggsave("figs/all_mics.svg", height = 6, width = 6)

# ggplot(data = hist_dat) +
#   geom_bar(aes(x = logmic, y = count,
#            fill = type),
#            alpha = 0.6,
#            stat = "identity",
#            width = 0.05) +
#   # here are labels on top of higher bars:
#   geom_text(aes(x = logmic, y = count, label = lab,
#                 hjust = hjustt, vjust = vjustt),
#             data = to_lab,
#              nudge_y = to_lab$nudge_yy,
#             size = 2.5) +
#   # abandoning
#   facet_wrap(~ type, ncol = 1, scales = "free_y") +
#   scale_x_continuous(breaks = log2(x_breaks),
#                      labels = x_breaks,
#                      limits = log2(xlim)) +
```

```
#   scale_fill_manual(values = pal[2:3]) +
#   ylab("") +
#   xlab("MIC90") +
#   theme_bw() +
#   theme(legend.position = "none")
```

Combine the two figures above for the journal on common axes

```
X_LOWER <- 1965

# here comes the model again:
newdata <- data.frame(isolate_yr = seq(X_LOWER, max(dat_nos_only$isolate_yr) + 2))
ci <- predict(mod_weighted_nos_only,
                newdata = newdata,
                interval = "confidence") %>%
  as.data.frame() %>%
  mutate(lwr = ifelse(lwr < Y_LOWER, Y_LOWER, lwr),
         isolate_yr = seq(X_LOWER, max(dat_nos_only$isolate_yr) + 2))

mutate(ci, diff = upr - lwr) %>% arrange(diff)
```

```
##          fit       lwr       upr isolate_yr      diff
## 36 -6.134709 -6.300709 -5.968709       2000 0.3319998
## 35 -6.145052 -6.311374 -5.978730       1999 0.3326445
## 37 -6.124366 -6.291139 -5.957592       2001 0.3335467
## 34 -6.155395 -6.323129 -5.987661       1998 0.3354680
## 38 -6.114023 -6.282650 -5.945395       2002 0.3372549
## 33 -6.165738 -6.335946 -5.995530       1997 0.3404162
## 39 -6.103680 -6.275207 -5.932152       2003 0.3430544
## 32 -6.176081 -6.349780 -6.002382       1996 0.3473982
## 40 -6.093337 -6.268757 -5.917916       2004 0.3508415
## 31 -6.186424 -6.364571 -6.008277       1995 0.3562947
## 41 -6.082994 -6.263237 -5.902750       2005 0.3604873
## 30 -6.196767 -6.380250 -6.013284       1994 0.3669662
## 42 -6.072650 -6.258574 -5.886727       2006 0.3718474
## 29 -6.207110 -6.396742 -6.017479       1993 0.3792631
## 43 -6.062307 -6.254692 -5.869923       2007 0.3847698
## 28 -6.217453 -6.413970 -6.020937       1992 0.3930327
## 44 -6.051964 -6.251516 -5.852413       2008 0.3991029
## 27 -6.227796 -6.431859 -6.023733       1991 0.4081261
## 45 -6.041621 -6.248972 -5.834271       2009 0.4147004
## 26 -6.238139 -6.450340 -6.025938       1990 0.4244021
## 46 -6.031278 -6.246991 -5.815566       2010 0.4314251
## 25 -6.248482 -6.469347 -6.027617       1989 0.4417299
## 47 -6.020935 -6.245511 -5.796360       2011 0.4491512
## 24 -6.258825 -6.488821 -6.028830       1988 0.4599907
## 48 -6.010592 -6.244475 -5.776710       2012 0.4677649
## 23 -6.269168 -6.508707 -6.029630       1987 0.4790778
## 49 -6.000249 -6.243831 -5.756667       2013 0.4871644
## 22 -6.279512 -6.528960 -6.030063       1986 0.4988964
## 50 -5.989906 -6.243536 -5.736276       2014 0.5072596
## 21 -6.289855 -6.549536 -6.030173       1985 0.5193627
## 51 -5.979563 -6.243549 -5.715578       2015 0.5279710
```

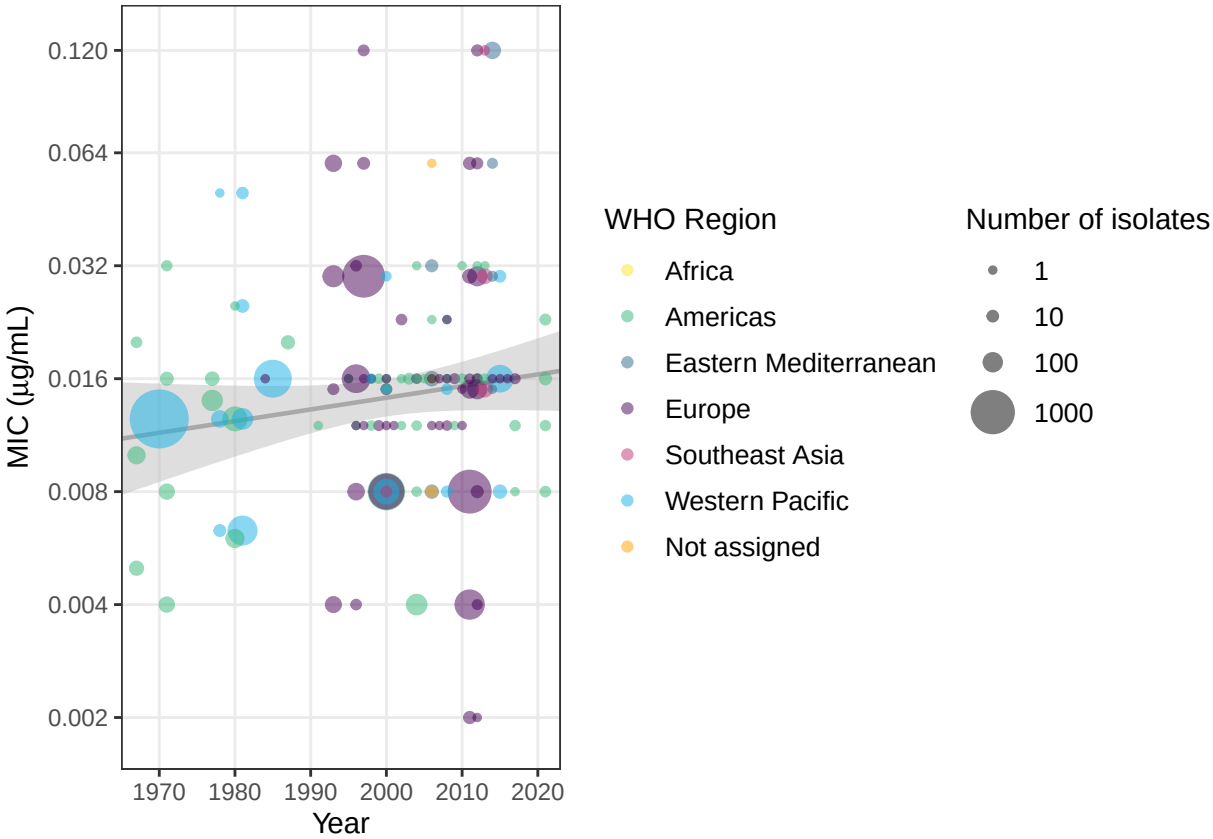
```
## 20 -6.300198 -6.570399 -6.029996       1984 0.5404032
## 52 -5.969220 -6.243834 -5.694605       2016 0.5492290
## 19 -6.310541 -6.591517 -6.029564       1983 0.5619534
## 53 -5.958877 -6.244363 -5.673391       2017 0.5709724
## 18 -6.320884 -6.612862 -6.028905       1982 0.5839568
## 54 -5.948534 -6.245108 -5.651960       2018 0.5931479
## 17 -6.331227 -6.634409 -6.028045       1981 0.6063642
## 55 -5.938191 -6.246045 -5.630336       2019 0.6157089
## 16 -6.341570 -6.656136 -6.027004       1980 0.6291323
## 56 -5.927848 -6.247155 -5.608541       2020 0.6386144
## 15 -6.351913 -6.678025 -6.025801       1979 0.6522233
## 57 -5.917505 -6.248419 -5.586590       2021 0.6618288
## 14 -6.362256 -6.700058 -6.024454       1978 0.6756043
## 58 -5.907162 -6.249822 -5.564501       2022 0.6853205
## 13 -6.372599 -6.722222 -6.022976       1977 0.6992460
## 59 -5.896819 -6.251350 -5.542288       2023 0.7090621
## 12 -6.382942 -6.744503 -6.021381       1976 0.7231229
## 11 -6.393285 -6.766891 -6.019679       1975 0.7472124
## 10 -6.403628 -6.789376 -6.017881       1974 0.7714947
## 9  -6.413971 -6.811947 -6.015995       1973 0.7959522
## 8  -6.424314 -6.834599 -6.014030       1972 0.8205690
## 7  -6.434657 -6.857323 -6.011992       1971 0.8453314
## 6  -6.445000 -6.880114 -6.009887       1970 0.8702268
## 5  -6.455343 -6.902966 -6.007721       1969 0.8952442
## 4  -6.465686 -6.925873 -6.005500       1968 0.9203737
## 3  -6.476029 -6.948833 -6.003226       1967 0.9456063
## 2  -6.486373 -6.971840 -6.000906       1966 0.9709340
## 1  -6.496716 -6.994890 -5.998541       1965 0.9963494
```

```
p1 <- ggplot(dat_nos_only %>%
             filter(!is.na(micnos_1)) %>%
             group_by(isolate_yr, micnos_1, who_name) %>%
             summarise(count = sum(count)) %>%
             mutate(who_name = factor(who_name, levels = c("Africa",
                                                           "Americas",
                                                           "Eastern Mediterranean",
                                                           "Europe",
                                                           "Southeast Asia",
                                                           "Western Pacific",
                                                           "Not assigned"))) %>%
           arrange(desc(count))) +
  geom_ribbon(data = ci, aes(ymin = 2**lwr, ymax = 2**upr, x = isolate_yr),
              alpha = 0.5, fill = "grey", colour = NA) +
  geom_line(data = ci, aes(x = isolate_yr, y = 2**fit),
            colour = "darkgrey", linewidth = 0.8) +
  geom_point(aes(x = isolate_yr, y = micnos_1,
                 col = who_name,
                 size = count), alpha = 0.5,
             show.legend = TRUE) +
  scale_color_manual(name = "WHO Region", values = pal, drop = FALSE) +
  scale_size_continuous(range = c(1,10), name = "Number of isolates",
                        breaks = c(1, 10, 100, 1000), limits = SIZE_LIMS) +
  scale_y_continuous(breaks = yticks, minor_breaks = NULL) +
  scale_x_continuous(limits = c(X_LOWER, 2023), expand = c(0,0),
```

```
                     breaks = seq(1970, 2020, 10), minor_breaks = NULL) +
  labs(x = "Year", y = expression("MIC (" * mu * "g/mL)")) +
  coord_trans(y = "log2", ylim = YLIM) +
  theme_bw() +
  theme(
    legend.text=element_text(size=10),
    legend.box = "horizontal",
    legend.box.just = "top",
    legend.spacing.x = unit(0, "cm"),
    legend.background = element_rect(fill = NA),
    plot.background = element_rect(fill = "transparent", colour = "transparent")
  ) +
  guides(size = guide_legend(order = 2), col = guide_legend(order = 1))
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by isolate_yr, micnos_1, and who_name.
## i Output is grouped by isolate_yr and micnos_1.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(isolate_yr, micnos_1, who_name))` for per-operation
##   grouping (`?dplyr::dplyr_by`) instead.
```

```
p1
```




```
X_LOWER <- 1927

# here comes the model again:
newdata <- data.frame(isolate_yr = seq(X_LOWER, max(dat_agg$isolate_yr) + 2))
ci <- predict(mod_weighted_agg,
```

```
                newdata = newdata,
                interval = "confidence") %>%
  as.data.frame() %>%
  mutate(lwr = ifelse(lwr < Y_LOWER, Y_LOWER, lwr),
         isolate_yr = seq(X_LOWER, max(dat_agg$isolate_yr) + 2))

mutate(ci, diff = upr - lwr) %>% arrange(diff)
```

```
##          fit       lwr       upr isolate_yr      diff
## 73 -5.962816 -6.101202 -5.824430       1999 0.2767720
## 72 -5.962281 -6.101180 -5.823381       1998 0.2778000
## 74 -5.963351 -6.102257 -5.824446       2000 0.2778116
## 71 -5.961745 -6.102181 -5.821309       1997 0.2808728
## 75 -5.963887 -6.104335 -5.823439       2001 0.2808959
## 70 -5.961210 -6.104172 -5.818247       1996 0.2859245
## 76 -5.964422 -6.107402 -5.821443       2002 0.2859586
## 69 -5.960674 -6.107100 -5.814248       1995 0.2928528
## 77 -5.964958 -6.111407 -5.818509       2003 0.2928971
## 68 -5.960139 -6.110903 -5.809374       1994 0.3015283
## 78 -5.965493 -6.116285 -5.814702       2004 0.3015821
## 67 -5.959603 -6.115506 -5.803700       1993 0.3118052
## 79 -5.966029 -6.121963 -5.810095       2005 0.3118677
## 66 -5.959068 -6.120833 -5.797302       1992 0.3235310
## 80 -5.966564 -6.128365 -5.804764       2006 0.3236013
## 65 -5.958532 -6.126809 -5.790255       1991 0.3365542
## 81 -5.967100 -6.135416 -5.798784       2007 0.3366314
## 64 -5.957997 -6.133362 -5.782631       1990 0.3507304
## 82 -5.967635 -6.143042 -5.792229       2008 0.3508137
## 63 -5.957461 -6.140424 -5.774498       1989 0.3659256
## 83 -5.968171 -6.151178 -5.785164       2009 0.3660143
## 62 -5.956926 -6.147935 -5.765917       1988 0.3820181
## 84 -5.968706 -6.159762 -5.777651       2010 0.3821116
## 61 -5.956390 -6.155840 -5.756940       1987 0.3988994
## 85 -5.969242 -6.168740 -5.769743       2011 0.3989971
## 60 -5.955855 -6.164091 -5.747618       1986 0.4164737
## 86 -5.969777 -6.178065 -5.761490       2012 0.4165750
## 59 -5.955319 -6.172648 -5.737991       1985 0.4346568
## 87 -5.970313 -6.187694 -5.752932       2013 0.4347613
## 58 -5.954784 -6.181471 -5.728096       1984 0.4533755
## 88 -5.970848 -6.197590 -5.744107       2014 0.4534829
## 57 -5.954248 -6.190531 -5.717965       1983 0.4725661
## 89 -5.971384 -6.207722 -5.735046       2015 0.4726760
## 56 -5.953713 -6.199799 -5.707626       1982 0.4921735
## 90 -5.971919 -6.218062 -5.725777       2016 0.4922857
## 55 -5.953177 -6.209252 -5.697102       1981 0.5121499
## 91 -5.972455 -6.228587 -5.716323       2017 0.5122639
## 54 -5.952642 -6.218868 -5.686415       1980 0.5324536
## 92 -5.972990 -6.239275 -5.706706       2018 0.5325694
## 53 -5.952106 -6.228630 -5.675582       1979 0.5530486
## 93 -5.973526 -6.250109 -5.696943       2019 0.5531660
## 52 -5.951571 -6.238522 -5.664619       1978 0.5739035
## 94 -5.974061 -6.261072 -5.687050       2020 0.5740223
## 51 -5.951035 -6.248531 -5.653540       1977 0.5949911
## 95 -5.974597 -6.272152 -5.677041       2021 0.5951112
```

```
## 50 -5.950500 -6.258643 -5.642356       1976 0.6162875
## 96 -5.975132 -6.283337 -5.666928       2022 0.6164086
## 49 -5.949964 -6.268850 -5.631078       1975 0.6377717
## 97 -5.975668 -6.294615 -5.656721       2023 0.6378938
## 48 -5.949429 -6.279141 -5.619716       1974 0.6594253
## 98 -5.976203 -6.305977 -5.646429       2024 0.6595484
## 47 -5.948893 -6.289509 -5.608277       1973 0.6812323
## 46 -5.948358 -6.299947 -5.596769       1972 0.7031783
## 45 -5.947822 -6.310448 -5.585197       1971 0.7252508
## 44 -5.947287 -6.321006 -5.573568       1970 0.7474384
## 43 -5.946751 -6.331617 -5.561886       1969 0.7697314
## 42 -5.946216 -6.342276 -5.550155       1968 0.7921207
## 41 -5.945680 -6.352979 -5.538381       1967 0.8145984
## 40 -5.945145 -6.363724 -5.526566       1966 0.8371575
## 39 -5.944609 -6.374505 -5.514714       1965 0.8597914
## 38 -5.944074 -6.385321 -5.502827       1964 0.8824945
## 37 -5.943538 -6.396169 -5.490908       1963 0.9052616
## 36 -5.943003 -6.407047 -5.478959       1962 0.9280879
## 35 -5.942467 -6.417952 -5.466983       1961 0.9509691
## 34 -5.941932 -6.428883 -5.454981       1960 0.9739015
## 33 -5.941396 -6.439837 -5.442956       1959 0.9968814
## 32 -5.940861 -6.450814 -5.430908       1958 1.0199057
## 31 -5.940325 -6.461811 -5.418840       1957 1.0429713
## 30 -5.939790 -6.472828 -5.406752       1956 1.0660757
## 29 -5.939254 -6.483863 -5.394646       1955 1.0892164
## 28 -5.938719 -6.494914 -5.382523       1954 1.1123911
## 27 -5.938183 -6.505982 -5.370385       1953 1.1355977
## 26 -5.937648 -6.517065 -5.358231       1952 1.1588343
## 25 -5.937112 -6.528162 -5.346063       1951 1.1820991
## 24 -5.936577 -6.539272 -5.333882       1950 1.2053906
## 23 -5.936041 -6.550395 -5.321688       1949 1.2287071
## 22 -5.935506 -6.561530 -5.309482       1948 1.2520473
## 21 -5.934970 -6.572675 -5.297266       1947 1.2754098
## 20 -5.934435 -6.583832 -5.285038       1946 1.2987936
## 19 -5.933899 -6.594998 -5.272801       1945 1.3221974
## 18 -5.933364 -6.606174 -5.260554       1944 1.3456202
## 17 -5.932828 -6.617359 -5.248298       1943 1.3690610
## 16 -5.932293 -6.628553 -5.236033       1942 1.3925190
## 15 -5.931758 -6.639754 -5.223761       1941 1.4159932
## 14 -5.931222 -6.650963 -5.211481       1940 1.4394829
## 13 -5.930687 -6.662180 -5.199193       1939 1.4629874
## 12 -5.930151 -6.673404 -5.186898       1938 1.4865058
## 11 -5.929616 -6.684634 -5.174597       1937 1.5100377
## 10 -5.929080 -6.695871 -5.162289       1936 1.5335823
## 9  -5.928545 -6.707114 -5.149975       1935 1.5571390
## 8  -5.928009 -6.718363 -5.137655       1934 1.5807074
## 7  -5.927474 -6.729617 -5.125330       1933 1.6042870
## 6  -5.926938 -6.740877 -5.113000       1932 1.6278771
## 5  -5.926403 -6.752141 -5.100664       1931 1.6514775
## 4  -5.925867 -6.763411 -5.088323       1930 1.6750876
## 3  -5.925332 -6.774685 -5.075978       1929 1.6987070
## 2  -5.924796 -6.785964 -5.063628       1928 1.7223354
## 1  -5.924261 -6.797247 -5.051274       1927 1.7459724
```

```
p2 <- ggplot(dat_agg %>%
             filter(!is.na(mic_90_agg)) %>%
             group_by(isolate_yr, mic_90_agg, who_name) %>%
             summarise(count = sum(count)) %>%
             mutate(who_name = factor(who_name, levels = c("Africa",
                                                           "Americas",
                                                           "Eastern Mediterranean",
                                                           "Europe",
                                                           "Southeast Asia",
                                                           "Western Pacific",
                                                           "Not assigned"))) %>%
           arrange(desc(count))) +
  geom_ribbon(data = ci, aes(ymin = 2**lwr, ymax = 2**upr, x = isolate_yr),
              alpha = 0.5, fill = "grey", colour = NA) +
  geom_line(data = ci, aes(x = isolate_yr, y = 2**fit),
            colour = "darkgrey", linewidth = 0.8) +
  geom_point(aes(x = isolate_yr, y = mic_90_agg,
                 col = who_name,
                 size = count), alpha = 0.5) +
  scale_color_discrete(name = "WHO Region",
                       type = pal,
                       guide = guide_legend(override.aes = list(size = 5),
                                            ncol = 2),
                       drop = FALSE) +
  scale_size_continuous(range = c(1,10), name = "Number of isolates",
                        breaks = c(1, 10, 100, 1000), limits = SIZE_LIMS) +
  scale_y_continuous(breaks = yticks, minor_breaks = NULL) +
  scale_x_continuous(limits = c(X_LOWER, 2023), expand = c(0,0),
                     breaks = seq(1930, 2020, 10), minor_breaks = NULL) +
  xlab("Year") +
  ylab(expression("MIC90 (" * mu * "g/mL)")) +
  coord_trans(y = "log2", ylim = YLIM) +
  theme_bw() #+
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by isolate_yr, mic_90_agg, and who_name.
## i Output is grouped by isolate_yr and mic_90_agg.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(isolate_yr, mic_90_agg, who_name))` for per-operation
##   grouping (`?dplyr::dplyr_by`) instead.
```

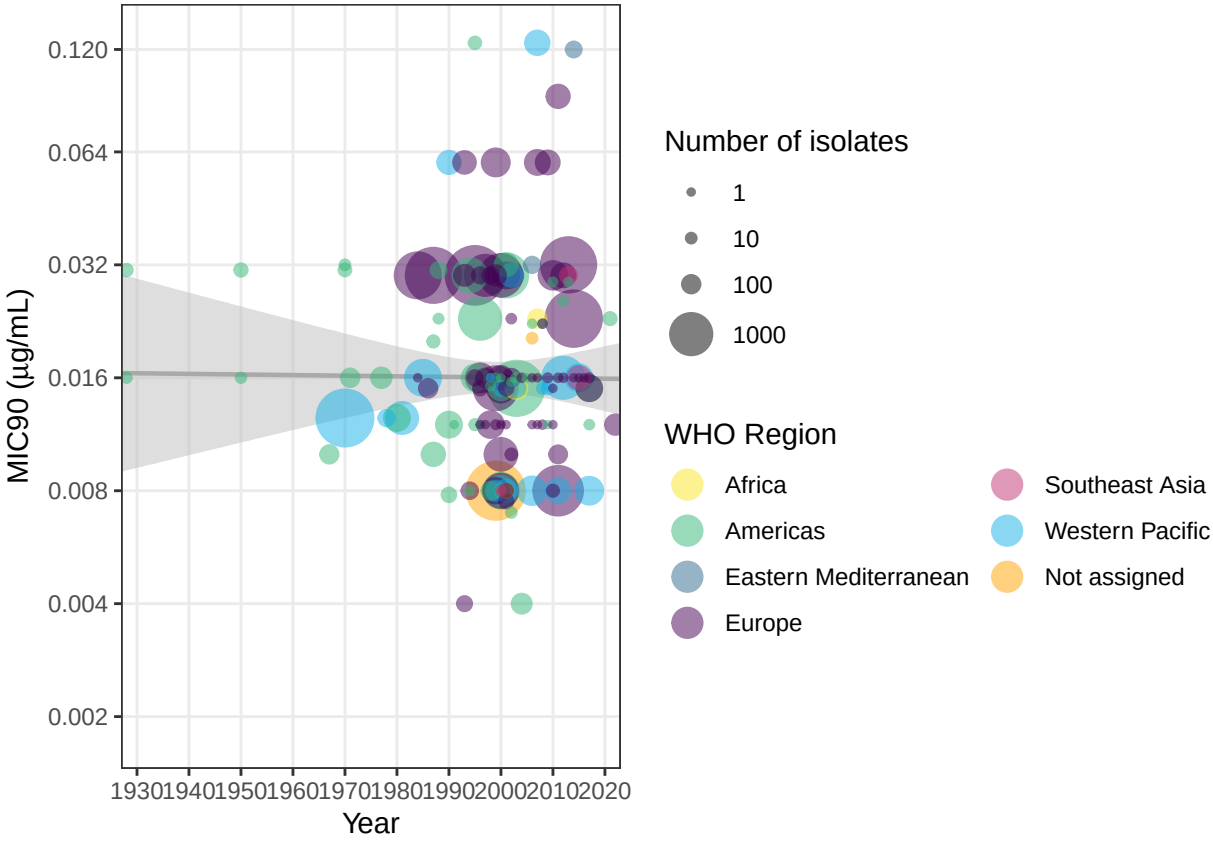
```
  # theme(
  #   legend.box = "horizontal",
  #   legend.box.just = "top",
  #   plot.background = element_rect(fill = "transparent", colour = "transparent")
  # )

p2
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_ribbon()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_line()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```




```
# This snippet to show all data in (d), including MIC90s of MICs in (c)
# hist_dat <- bind_rows(
#   dat_nos_only %>%
#     rename(value = micnos_1) %>%
#     summarise(count = sum(count), .by = c(value)) %>%
#     mutate(type = "mic_nos"),
#   dat_agg %>%
#     rename(value = mic_90_agg) %>%
#     summarise(count = sum(count), .by = c(value)) %>%
#     mutate(type = "mic90")) %>%
#   mutate(logmic = log2(value))
#
# to_lab <- hist_dat %>%
#   # only highest ones:
#   filter((type == "mic90" & count > 5000) |
#            (type == "mic_nos" & count > 1000)) %>%
#   mutate(nudge_yy = ifelse(type == "mic_nos",
#                            1200, 300),
#          countstr = format(count, big.mark = ","),
#          mic = ifelse(type == "mic_nos", "MIC", "MIC90"),
#          hjustt = ifelse(count %in% c(2727, 2100, 3097), 0, 0.5),
#          vjustt = ifelse(count %in% c(2100), 0, 0.5),
#          lab = ifelse(count %in% c(2727, 3097, 2030),
#                       paste0(mic, ":\n", value, "\n(n = ", countstr, ")"),
```

```
#                       paste0(mic, ": ", value, "\n(n = ", countstr, ")")))

hist_dat <- dat_raw %>% # from Rhys' aggregate file
  filter(!is.na(mic90_1)) %>%
  bind_rows(dat_nos_only) %>% # NOS only
  pivot_longer(cols = c(micnos_1, mic90_1), names_to = "type") %>%
  filter(!is.na(value)) %>%
  mutate(value = round(value, digits = 4)) %>%
  summarise(count = sum(count), .by = c(value, type)) %>%
  mutate(logmic = log2(value)) %>%
  mutate(type = ifelse(type == "mic90_1",
                       "mic90",
                       "mic_nos"))

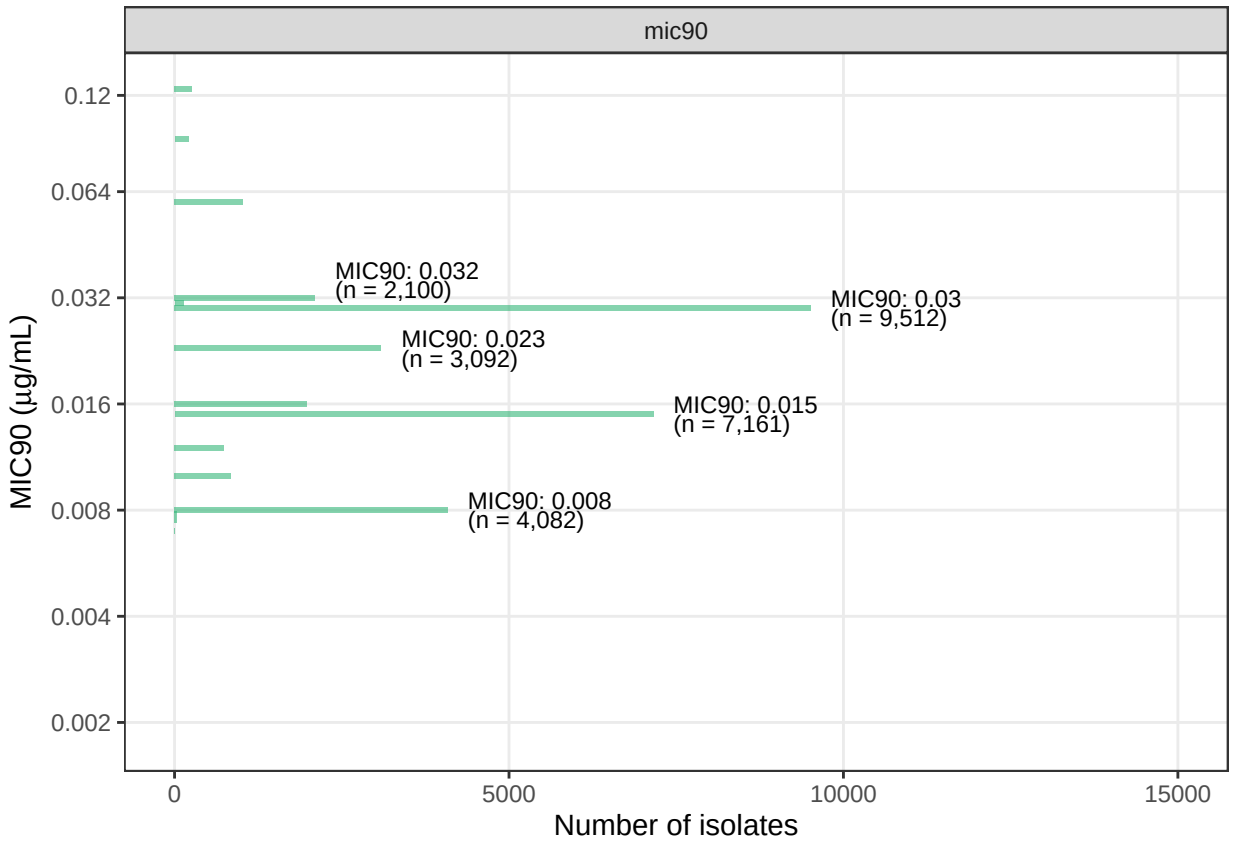
# for labels on bars:
to_lab <- hist_dat %>%
  # only highest ones:
  filter((type == "mic90" & count > 2000) |
           (type == "mic_nos" & count > 1000)) %>%
  mutate(nudge_yy = ifelse(type == "mic_nos",
                           1200, 300),
         countstr = format(count, big.mark = ","),
         mic = ifelse(type == "mic_nos", "MIC", "MIC90"),
         hjustt = ifelse(count %in% c(2727, 2100, 3097), 0, 0.5),
         vjustt = ifelse(count %in% c(2100), 0, 0.5),
         lab = ifelse(count %in% c(2727, 3097, 2030),
                      paste0(mic, ":\n", value, "\n(n = ", countstr, ")"),
                      paste0(mic, ": ", value, "\n(n = ", countstr, ")")))

x_breaks = c(0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.12)


p3 <- ggplot(data = hist_dat %>%
               filter(type == "mic90")) +
  geom_bar(aes(x = logmic, y = count,
           fill = type),
           alpha = 0.6,
           stat = "identity",
           width = 0.05) +
  # here are labels on top of higher bars:
  geom_text(aes(x = logmic, y = count, label = lab, vjust = vjustt),
            data = to_lab %>%
              filter(type == "mic90"),
             nudge_y = 300,
            lineheight = .7,
            hjust = 0,
            size = 3) +
  facet_wrap(~ type, ncol = 1, scales = "free_y") +
  scale_x_continuous(breaks = log2(x_breaks),
                     labels = x_breaks,
                     limits = log2(YLIM),
                     minor_breaks = NULL) +
  scale_y_continuous(minor_breaks = NULL, limits = c(0, 15000)) +
```

```
  scale_fill_manual(values = pal[2]) +
  ylab("Number of isolates") +
  xlab(expression("MIC90 (" * mu * "g/mL)")) +
  theme_bw() +
  theme(legend.position = "none") +
  coord_flip()

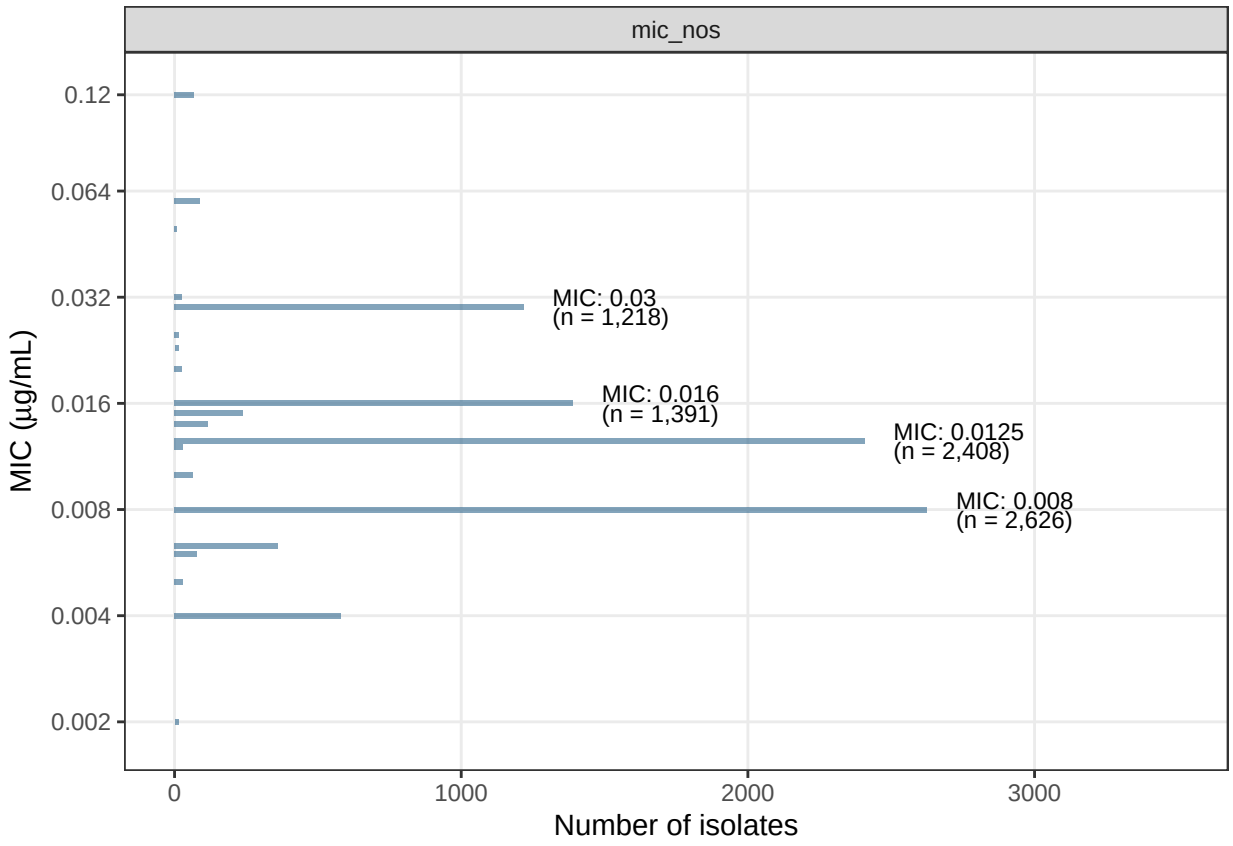
p3
```




```
p4 <- ggplot(data = hist_dat %>%
               filter(type == "mic_nos")) +
  geom_bar(aes(x = logmic, y = count,
           fill = type),
           alpha = 0.6,
           stat = "identity",
           width = 0.05) +
  # here are labels on top of higher bars:
  geom_text(aes(x = logmic, y = count, label = lab),
            data = to_lab %>%
               filter(type == "mic_nos"),
             nudge_y = 100,
            hjust = 0,
            lineheight = .7,
            #vjust = 1,
            #angle = -90,
            size = 3) +
  facet_wrap(~ type, ncol = 1, scales = "free_y") +
```

```
  scale_x_continuous(breaks = log2(x_breaks),
                     labels = x_breaks,
                     limits = log2(YLIM),
                    minor_breaks = NULL) +
  scale_y_continuous(minor_breaks = NULL, limits = c(0, 3500)) +
  scale_fill_manual(values = pal[3]) +
  ylab("Number of isolates") +
  xlab(expression("MIC (" * mu * "g/mL)")) +
  theme_bw() +
  theme(legend.position = "none") +
  coord_flip()
```

```
p4
```




```
p <- plot_grid(
          p1 + theme(legend.position = c(-0.55, 0.5),
                     plot.margin = margin(t = 20, r = 0, b = 0, l = 230,#l = 215,
                                          unit = "pt"),
                     legend.spacing.x = unit(0, "cm")),
          p2 + theme(legend.position = "none",
                     plot.margin = margin(t = 20, r = 0, b = 0, l = 45, #l = 5,
                                          unit = "pt")),
          labels = c("(a) Trend based on individual MICs",
                     "(b) Trend based on summarised MICs"),
          label_fontface = "plain",
          nrow = 2,
          label_x = c(0.45, 0.07),
```

```
          hjust = 0, label_size = 12)
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_ribbon()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_line()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```

```
p <- plot_grid(p,
      plot_grid(p4 + theme(plot.margin = margin(t = 20, r = 5, b = 0, l = 10,
                                        unit = "pt"),
                           strip.background = element_blank(),
                           strip.text = element_blank()),
                p3 + theme(plot.margin = margin(t = 20, r = 5, b = 0, l = 10,
                                        unit = "pt"),
                           strip.background = element_blank(),
                           strip.text = element_blank()), ncol = 1,
                labels = c("(c) Counts of individual MICs",
                           "(d) Counts of summarised MICs"),
          label_fontface = "plain",
          label_x = 0.1,
          hjust = 0, label_size = 12),
      ncol = 2,
               rel_widths = c(1, 0.45), align = "v")

HEIGHT <- 6.2
ggsave("figs/mic_trends_and_efforts.png", p, height = HEIGHT, width = HEIGHT * 1.8, bg = "white")
ggsave("figs/mic_trends_and_efforts.svg", p, height = HEIGHT, width = HEIGHT * 1.8, bg = "white")

# ggdraw(p) +
#   draw_label("Number of isolates", x = 0, y = 0.5, angle = 90,
#              vjust = 1.5, size = 12)
```

Ordinal regression

What happens when we try to reapply the linear regression treating MICs as ordinal? We could really treat MICs as categories as intermediate MICs are not (usually) tested. For this reason, we've had a look at ordinal regression in the chunk below. We resolved not to do this because the data was all created under different settings with (probably) different dilutions, different methods of calculating MIC90, etc.

First let's have a look at raw MICs only:

```
library(MASS)
```

```
##
## Attaching package: 'MASS'
```

```
## The following object is masked from 'package:dplyr':
##
##     select
```

```
ordinal_regn_please <- function(dat, bins, ylab,
                                weight = TRUE,
                                xlower = 1927,
```

```
                                xupper = 2024){
dat <- mutate(dat,
              mic_binned = cut(mic,
                               breaks = c(0, bins),
                               ordered_result = TRUE))

# two poss models:
if (weight){
  model <- polr(mic_binned ~ isolate_yr,
                data = dat,
                Hess = TRUE,
                weights = count, # here, weights are frequencies
                method = "probit") # fails to converge under logit: fatter tail
} else {
  message("here")
  model <- polr(mic_binned ~ isolate_yr,
                data = dat %>% uncount(count),
                Hess = TRUE)
}

# Fit the model - no need to log: that would be for visualisation purposes only under ordinal model
# tried using wts from logged model ......... not thinking too hard about that but we should at some
# it looks like weights need to be integers as we end up assuming binomial glm for ordinal *logistic*
# haven't thought too hard about this ... may need to be probit link

print(summary(model))
# note the slope here

new_data = data.frame(isolate_yr = seq(xlower + 2,
                                       xupper - 2))
preds <- predict(model,
                 newdata = new_data,
                 type = "probs") %>%
  as.data.frame() %>%
  mutate(max_probs = max.col(across(starts_with("(")))) %>%
  mutate(isolate_yr = new_data$isolate_yr)

if (length(unique(preds$max_probs)) == 1){
  message(paste0("Most probable class stays the same:",
                 unique(preds$max_probs)))
} else {
  message(paste0(c("Most probable class changes:\n", unique(preds$max_probs)),
                 collapse = " "))
}


preds_long <- pivot_longer(preds, starts_with("(")) %>%
  mutate(lower = str_extract(name, "(?<=\\()[^,]+") %>% as.numeric(),
         upper = str_extract(name, "(?<=,)[^]]+") %>% as.numeric(),
         height = lower + (upper - lower)*value) %>%
  filter(lower != 0)

preds_change <- preds %>%
```

```
    group_by(max_probs) %>%
    summarise(xmin = first(isolate_yr) - 0.5,
              xmax = last(isolate_yr) + 0.5) %>%
    mutate(ymin = bins[max_probs - 1],
           ymax = bins[max_probs])

  # so now we have colour bars representing class probability between the lower
  # and upper extents of the bins ... which don't change much because there isn't
  # much change in MIC slope over time
  ggplot(data = dat) +
    geom_point(aes(x = isolate_yr, y = mic, size = count),
               alpha = 0.5, col = "grey30") +
    scale_y_continuous(trans = "log2", breaks = bins, minor_breaks = NULL) +
    geom_ribbon(data = preds_long,
                aes(ymin = lower, ymax = height, fill = name, x = isolate_yr),
                alpha = 0.5) +
    scale_fill_discrete("MIC interval") +
    scale_size_continuous(range = c(1,10), name = "Number of\nisolates",
                          breaks = c(1, 10, 100, 1000), limits = SIZE_LIMS) +
    # (the -1 is because I removed the bottom bin)
    # this is a bit gross but I've put the line in the middle
    # geom_hline(yintercept = 2**(sum(log2(bins[preds$max_probs[1] + c(0,-1)])) / 2)) +
    # annotate(geom = "rect",
    #          xmin = xlower,
    #           xmax = max(new_data$isolate_yr),
    #           ymin = bins[preds$max_probs[1] - 1],
    #           ymax = bins[preds$max_probs[1]], col = "black", fill = NA) +
    geom_rect(data = preds_change,
              aes(xmin = xmin, xmax = xmax, ymin = ymin, ymax = ymax),
              col = "black", fill = NA) +
    scale_x_continuous(breaks = seq(ceiling(xlower / 10) * 10,
                                    floor(xupper / 10) * 10, 10),
                       minor_breaks = NULL, expand = FALSE,
                       limits = c(xlower - 0.1, xupper + 0.1)) +
    xlab("Year") +
    ylab(ylab) +
    theme_bw() +
    theme(
      legend.box = "horizontal",
      legend.box.just = "top",
      plot.background = element_rect(fill = "transparent", colour = "transparent"),
      plot.title = element_text(size=11)
    )
}

bins = c(0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256)
# bins = 2**(log2(bins) - 0.5)

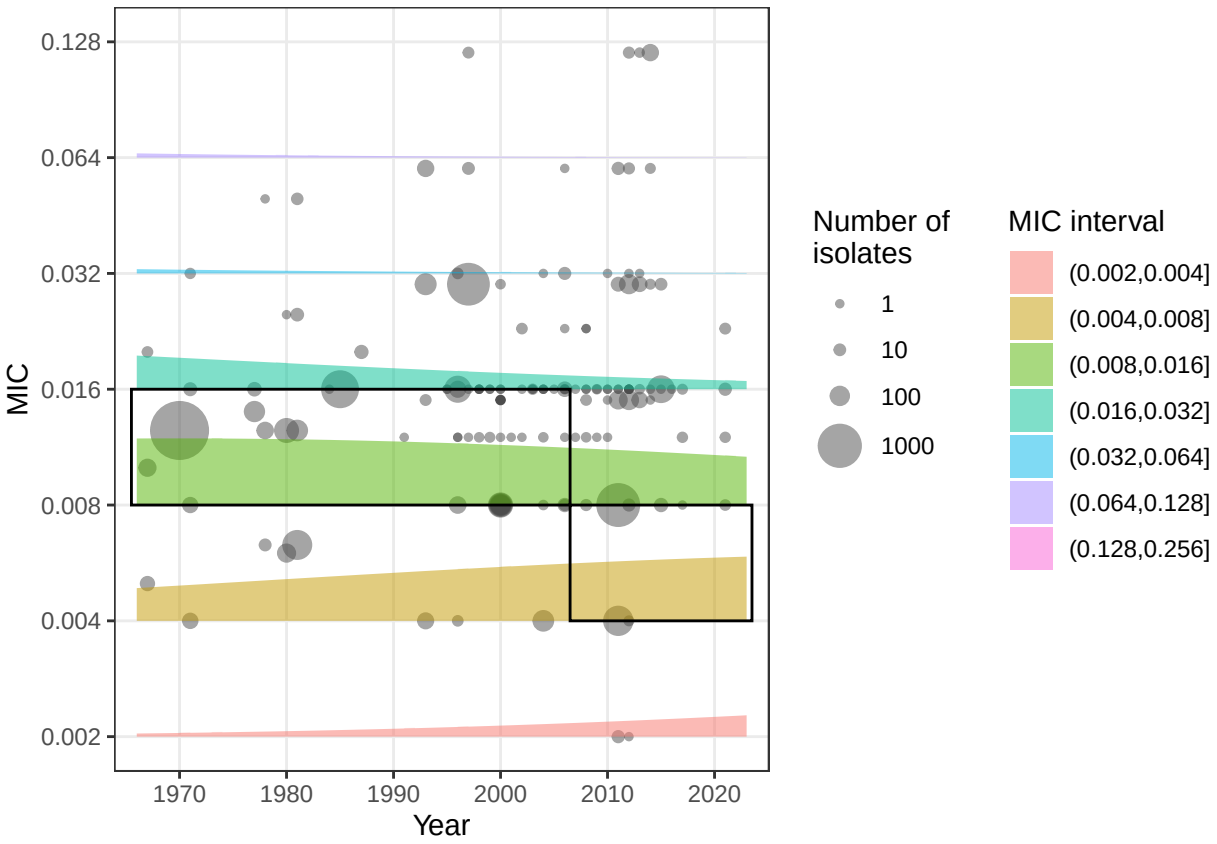
# that's interesting .. this slips down a class!
p1 <- ordinal_regn_please(dat_nos_only %>%
                            mutate(mic = micnos_1),
                          # bins = c(0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128),
                          weight = TRUE,
```

```
                    bins = bins, xupper = 2025,
                    xlower = 1964, ylab = "MIC")
```

```
## Call:
## polr(formula = mic_binned ~ isolate_yr, data = dat, weights = count,
##     Hess = TRUE, method = "probit")
##
## Coefficients:
##                Value Std. Error t value
## isolate_yr -0.01754  1.082e-05   -1620
##
## Intercepts:
##                                Value          Std. Error     t value
## (0,0.002]|(0.002,0.004]     -3.806530e+01  4.000000e-04 -9.224765e+04
## (0.002,0.004]|(0.004,0.008] -3.655150e+01  2.000000e-03 -1.818789e+04
## (0.004,0.008]|(0.008,0.016] -3.519760e+01  2.240000e-02 -1.568382e+03
## (0.008,0.016]|(0.016,0.032] -3.387900e+01  2.570000e-02 -1.318738e+03
## (0.016,0.032]|(0.032,0.064] -3.284970e+01  3.570000e-02 -9.196834e+02
## (0.032,0.064]|(0.064,0.128] -3.253310e+01  3.590000e-02 -9.068079e+02
## (0.064,0.128]|(0.128,0.256]  2.068948e+12  3.590000e-02  5.766852e+13
##
## Residual Deviance: 23170.82
## AIC: 23186.82
```

```
## Most probable class changes:
##  4 3
```

```
p1
```

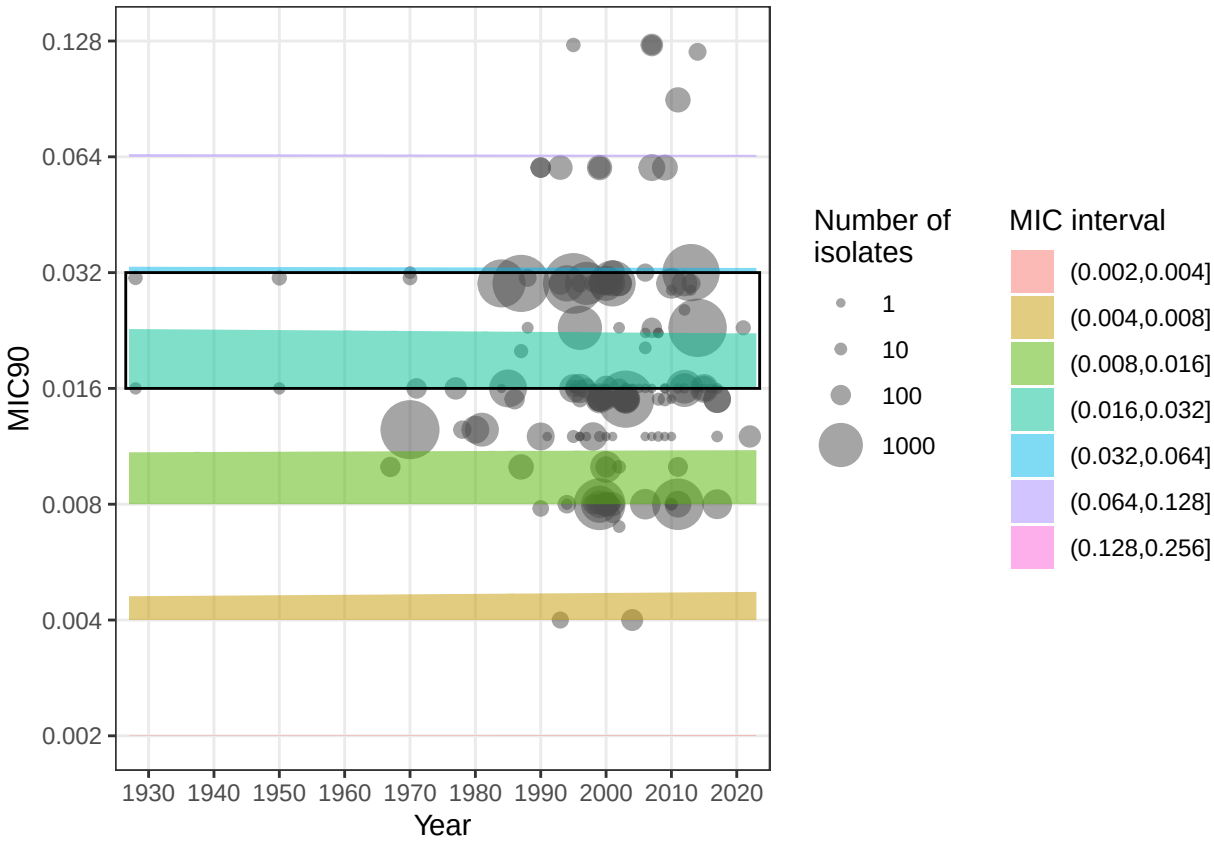


```r
p2 <- ordinal_regn_please(dat_agg %>% mutate(mic = mic_90_agg),
                    bins = bins,
                    xlower = 1925, xupper = 2025,
                    ylab = "MIC90")
```

```
## Call:
## polr(formula = mic_binned ~ isolate_yr, data = dat, weights = count,
##     Hess = TRUE, method = "probit")
##
## Coefficients:
##                Value Std. Error t value
## isolate_yr -0.001305  1.298e-05  -100.5
##
## Intercepts:
##                               Value       Std. Error  t value
## (0,0.002]|(0.002,0.004]         -7.2184      0.0007 -10619.5774
## (0.002,0.004]|(0.004,0.008]     -5.2314      0.0034  -1543.1434
## (0.004,0.008]|(0.008,0.016]     -3.5253      0.0262   -134.7889
## (0.008,0.016]|(0.016,0.032]     -2.4632      0.0264    -93.4417
## (0.016,0.032]|(0.032,0.064]     -0.8936      0.0287    -31.1745
## (0.032,0.064]|(0.064,0.128]     -0.3850      0.0288    -13.3834
## (0.064,0.128]|(0.128,0.256]     29.7652      0.0288   1034.6071
##
## Residual Deviance: 99561.28
## AIC: 99577.28
```

```
## Most probable class stays the same:5
```

```r
p2
```



Caption: Ordinal regression of MIC data (as in Figure 4): coloured regions represent predicted probability of MIC interval over time; the most probable interval, which does not change over time in either case, is highlighted with a black rectangle. Intervals are right-inclusive: the interval (0.008, 0.016] includes MIC 0.016 and excludes MIC 0.008.

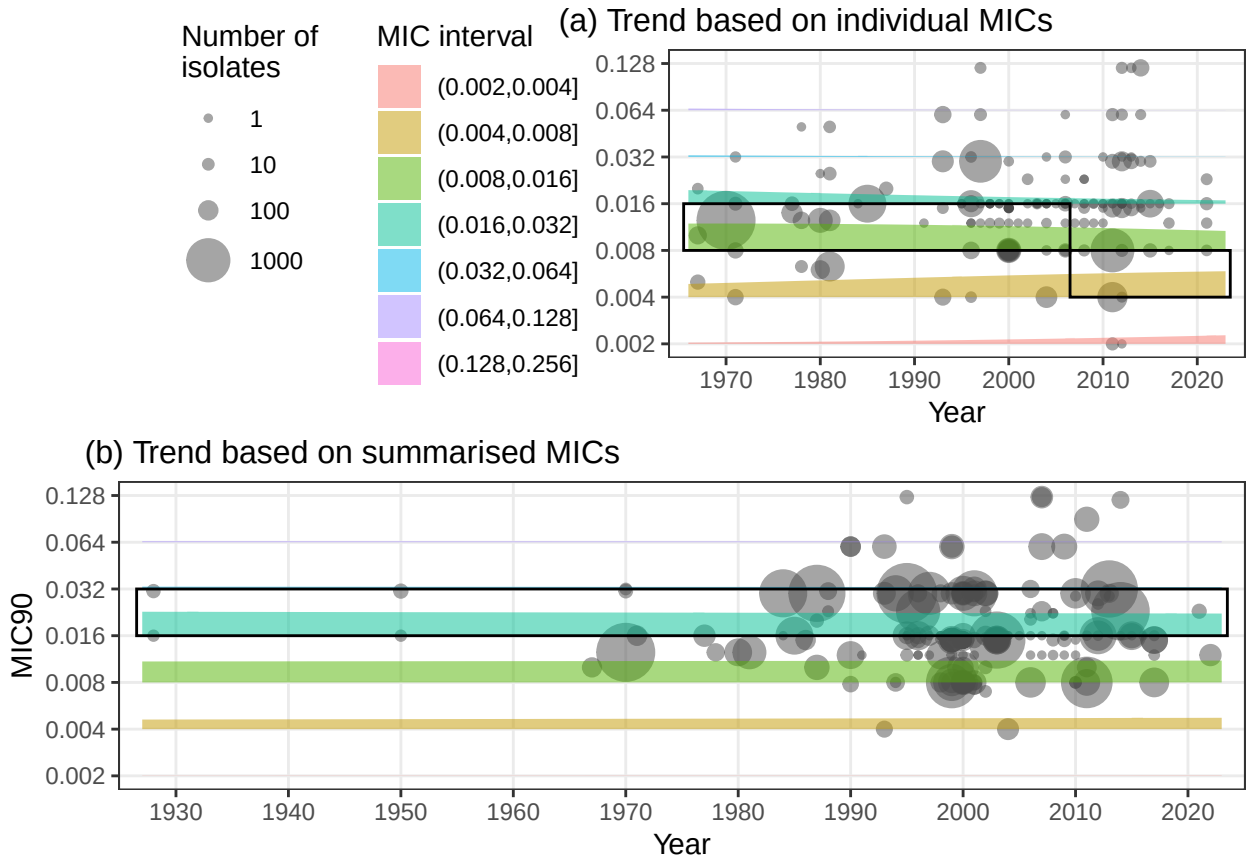
Here we put the subplots together:

```
# p <- plot_grid(p2 + theme(legend.position = "none"),
#           p1 + theme(legend.position = "none"),
#           rel_widths = c(0.6, 0.45))
#
# plot_grid(p, get_legend(p2), ncol = 1, rel_heights = c(0.8, 0.25))
#
# ggsave("figs/ordinal_regression.png", height = 7, width = 9, bg = "transparent")
# ggsave("figs/ordinal_regression.svg", height = 7, width = 9, bg = "transparent")


p <- plot_grid(
          p1 + theme(legend.position = c(-0.5, 0.5),
                     plot.margin = margin(t = 20, r = 0, b = 0, l = 212,
                                          unit = "pt")),
          p2 + theme(legend.position = "none",
                     plot.margin = margin(t = 20, r = 0, b = 0, l = 5,
                                          unit = "pt")),
          labels = c("(a) Trend based on individual MICs",
                     "(b) Trend based on summarised MICs"),
          label_fontface = "plain",
          nrow = 2,
```

```
          #label_x = c(0.07, 0.45),
          label_x = c(0.45, 0.07),
          hjust = 0, label_size = 12)

p
```




```
ggsave("figs/ordinal_regression.png", height = 9, width = 8, bg = "white")
ggsave("figs/ordinal_regression.svg", height = 9, width = 8, bg = "white")
```